Dr. A. Jha

MA203 Numerical Methods

Winter Semester 2026

30.03.2026

Programming Assigment 1

Programming Assignment must be completed alone and submitted via Google Classroom by 17.04.2026 at 23:59. ***Do not take help from Generative AI****; if found, it will lead to cancellation of the assignment.*

Question 1: [**System of Equations (3+3+4 Points)**]

The goal of this exercise is to study the numerical solution of linear systems arising from electrical networks and to compare the performance of direct and iterative methods. Consider an n-loop electrical circuit. Using Kirchhoff's Voltage Law (KVL), the system can be written as

$$\mathbf{A}\mathbf{i} = \mathbf{b},$$

where $\mathbf{i}$ is the vector of loop currents.
The matrix $\mathbf{A} \in \mathbb{R}^{n\times n}$, i.e., of the size $n \times n$ and vector $\mathbf{b} \in \mathbb{R}^n$ are defined as follows:

- $\mathbf{A}_{ii} = R + 2r, \quad i = 1, 2, \ldots, n,$
- $\mathbf{A}_{i,i+1} = \mathbf{A}_{i+1,i} = -r, \quad i = 1, 2, \ldots, n-1,$
- all other entries of A are zero,
- $\mathbf{b}_i = V, \quad i = 1, 2, \ldots, n.$

Take

$$R = 4, \quad r = 1, \quad V = 5.$$

a) Write a code to construct the matrix $\mathbf{A}$ and vector $\mathbf{b}$ for a given number of loops n.

b) Solve the system using Gaussian Elimination.

c) Implement the Jacobi method and Gauss-Seidel method to solve the system. Use the stopping criterion

$$\|\mathbf{i}^{(k+1)} - \mathbf{i}^{(k)}\|_\infty < 10^{-6}.$$

d) Consider $n = 5, 10, 20, 50$. For each value of n, report:

- the solution vector $\mathbf{i}$,
- the number of iterations required for convergence (Jacobi and Gauss–Seidel),
- the computational time for each method.

e) Comment on the following:

- How does the performance of Gaussian Elimination compare with the iterative methods as n increases?

- Which iterative method converges faster and why?
- How does the structure of the matrix $\mathbf{A}$ influence the convergence of the iterative methods?

Question 2: [**Power Method (5+5 Points)**]

The goal of this exercise is to study the behavior of the Power method and interpret its results in an engineering context.

In many engineering applications, eigenvalues and eigenvectors arise in the analysis of dynamical systems. For example, in structural vibrations, the dominant eigenvalue corresponds to the most significant mode of response.

a) Write a Python code to implement the Power method for computing the dominant eigenvalue and the corresponding eigenvector of a matrix.

Use a stopping criterion of the form

$$\frac{\|\mathbf{x}^{(k+1)} - \mathbf{x}^{(k)}\|_\infty}{\|\mathbf{x}^{(k)}\|_\infty} < 10^{-6},$$

and normalize the eigenvector at each iteration using the infinity norm.

b) Apply your code to the following matrices:

For each case, report:

- the dominant eigenvalue,
- the corresponding eigenvector,
- the number of iterations required for convergence.

i)

$$\mathbf{A} = \begin{bmatrix} 4 & 1 & 1 & 1 \\ 1 & 3 & -1 & 1 \\ 1 & -1 & 2 & 0 \\ 1 & 1 & 0 & 2 \end{bmatrix}, \qquad \mathbf{x}_0 = \begin{bmatrix} 1 & -2 & 0 & 3 \end{bmatrix}^\top.$$

ii)

$$\mathbf{A} = \begin{bmatrix} 1 & 3 & 4 \\ 0 & 2 & 1 \\ 0 & 0 & -2 \end{bmatrix}, \qquad \mathbf{x}_0 = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}^\top.$$

iii)

$$\mathbf{A} = \begin{bmatrix} 3 & 0 & 0 \\ -4 & 6 & 2 \\ 16 & -15 & -5 \end{bmatrix}, \qquad \mathbf{x}_0 = \begin{bmatrix} 0 & 0.5 & 0.25 \end{bmatrix}^\top.$$

c) Compute the eigenvalues of the matrices in **ii)** and **iii)** analytically. Compare with the numerical results and explain the observed behavior of the Power method in each case. In particular, comment on:

- the rate of convergence in relation to the ratio $\left|\frac{\lambda_2}{\lambda_1}\right|$,
- cases where convergence is slow or fails,
- the role of the initial vector $\mathbf{x}_0$.